

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

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Exposé VIII. The Finiteness Theorem

1. A Biduality Spectral Sequence¹

Let us state the result we wish to establish:

Proposition 1.1. *Let A be a Noetherian ring and let I be an ideal of A . Put $X = \operatorname{Spec}(A)$ and $Y = V(I)$. Let M be a finitely generated A -module of finite projective dimension. Let $\mathcal{F} = \widetilde{M}$ be the $\mathcal{O}_X$ -module associated with M .*

1) *There is a spectral sequence*

$$H_Y(X, \mathcal{F}) \Leftarrow \operatorname{Ext}_Y^p(\operatorname{Ext}^{-q}(M, A), A).$$

2) *There is a spectral sequence*

$$\mathcal{H}_Y(X, \mathcal{F}) \Leftarrow \operatorname{Ext}_Y^p(\operatorname{Ext}^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

Of course, 2) follows from 1) by observing that, if M and N are two finitely generated A -modules and if we put $\mathcal{F} = \widetilde{M}$ and $\mathcal{G} = \widetilde{N}$, then there are isomorphisms

$$\begin{aligned} \mathcal{H}_Y(\mathcal{F}) &\simeq \widetilde{H_Y(X, \mathcal{F})}, \\ \operatorname{Ext}_Y(\mathcal{F}, \mathcal{G}) &\simeq \widetilde{\operatorname{Ext}_Y(\mathcal{F}, \mathcal{G})}, \\ \operatorname{Ext}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) &\simeq \widetilde{\operatorname{Ext}_A(M, N)}. \end{aligned}$$

Let $\mathcal{C}$ be the category of A -modules, and let $\mathbf{Ab}$ be the category of abelian groups. Let $\underline{F}$ be the functor

$$\begin{aligned} \underline{F}: \mathcal{C} &\longrightarrow \mathbf{Ab}, \\ M &\longmapsto \Gamma_Y(\widetilde{M}). \end{aligned}$$

We know from Exposé II that there is an isomorphism of ∂ -functors

$$H_Y^*(X, \widetilde{M}) \simeq R^*\underline{F}(M).$$

Furthermore, let Ext_Y^* denote the right derived functors, in the second argument, of

$$\underline{F} \circ \operatorname{Hom}: \mathcal{C}^\circ \times \mathcal{C} \longrightarrow \mathbf{Ab}.$$

We know from Exposé VI that there is an isomorphism of ∂ -functors

$$\operatorname{Ext}_Y^*(M, N) \simeq \operatorname{Ext}_Y^*(\widetilde{M}, \widetilde{N}).$$

Finally, let us record the following result from Exposé VI: if C is an injective A -module and N is a finitely generated A -module, the sheaf

$$\mathcal{H}om(\widetilde{N}, \widetilde{C}) \simeq \widetilde{\operatorname{Hom}(N, C)}$$

is flasque; hence

$$R^1\underline{F}(\operatorname{Hom}(N, C)) = 0.$$

It remains for us to prove the following result.

¹A reader familiar with Verdier's language of derived categories will recognize the spectral sequence associated with a *biduality isomorphism*. Cf. SGA 6, I.

Lemma 1.2. — Let A be a Noetherian ring and let $\mathcal{C}$ be the category of A -modules. Let $\underline{F}: \mathcal{C} \rightarrow \mathbf{Ab}$ be a left exact additive functor such that, for every finitely generated A -module N and every injective A -module C , one has

$$R^1 \underline{F}(\mathrm{Hom}(N, C)) = 0.$$

Let M be a finitely generated A -module of finite projective dimension. There is a spectral sequence

$$R^* \underline{F}(M) \leftarrow \mathrm{Ext}_{\underline{F}}^p(\mathrm{Ext}^{-q}(M, A), A),$$

where $\mathrm{Ext}_{\underline{F}}^p$ denotes the p -th right derived functor of $\underline{F} \circ \mathrm{Hom}$.

We shall consider only complexes whose differential has degree $+1$. By the hypothesis on M , there exists a projective resolution of M of finite length,

$$u: L^\bullet \rightarrow M,$$

where, moreover, the L^p are finitely generated modules and $L^p = 0$ if $p \notin [-n, 0]$. On the other hand, let $v: M \rightarrow I^\bullet$ be an injective resolution of M . I claim that

$$v \circ u: L^\bullet \rightarrow I^\bullet \tag{1.1}$$

is an injective resolution of $L^\bullet$. We must specify what this means.

Definition 1.3. — Let $X^\bullet$ be a complex of A -modules. An *injective resolution* of $X^\bullet$ is a homomorphism of complexes

$$x: X^\bullet \rightarrow CX^\bullet$$

such that CX^p is injective for every $p \in \mathbb{Z}$, and x induces an isomorphism in homology.

Proposition 1.4. — Every left-bounded complex, i.e., every complex for which there exists $q \in \mathbb{Z}$ such that $X^p = 0$ for $p < q$, admits an injective resolution. Moreover, suppose that $u: X^\bullet \rightarrow Y^\bullet$ is a homomorphism of complexes (both left-bounded), and that $x: X^\bullet \rightarrow CX^\bullet$ and $y: Y^\bullet \rightarrow CY^\bullet$ are injective resolutions of $X^\bullet$ and $Y^\bullet$. Then there exists a homomorphism of complexes

$$Cu: CX^\bullet \rightarrow CY^\bullet,$$

unique up to homotopy, such that the diagram

$$\begin{array}{ccc} X^\bullet & \xrightarrow{x} & CX^\bullet \\ u \downarrow & & \downarrow Cu \\ Y^\bullet & \xrightarrow{y} & CY^\bullet \end{array}$$

commutes up to homotopy.

The proof is left to the reader.²

Let us recall a notation introduced in Exposé V.

Notation. — Let $X^\bullet$ and $Y^\bullet$ be two complexes. We denote by $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ the *simple complex* whose component of degree n is

$$(\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet))^n = \prod_{-p+q=n} \mathrm{Hom}(X^p, Y^q),$$

also denoted by $\mathrm{Hom}^n(X^\bullet, Y^\bullet)$, and whose differential is given by

$$\begin{aligned} \partial_n: \mathrm{Hom}^n(X^\bullet, Y^\bullet) &\rightarrow \mathrm{Hom}^{n+1}(X^\bullet, Y^\bullet), \\ \partial_n &= d' + (-1)^{n+1} d'', \end{aligned}$$

where d' and d'' are the differentials (of degree $+1$) induced by those of $X^\bullet$ and $Y^\bullet$.

Let $A^\bullet$ then be the complex defined by $A^p = 0$ if $p \neq 0$ and $A^0 = A$. Let

$$a: A^\bullet \rightarrow CA^\bullet$$

be an injective resolution of $A^\bullet$. Consider the double complex

$$Q^{p,q} = \mathrm{Hom}(\mathrm{Hom}(L^{-q}, A), CA^p). \tag{1.2}$$

The first spectral sequence of the bicomplex $\underline{F}Q^{\bullet,\bullet}$ will give the conclusion of Lemma 1.2.

²See also H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton Mathematical Series, vol. 19, Princeton University Press, 1956.

Set

$$L'^{\bullet} = \text{Hom}^{\bullet}(L^{\bullet}, A^{\bullet}), \quad (1.3)$$

and

$$P^{\bullet} = \text{Hom}^{\bullet}(L'^{\bullet}, CA^{\bullet}). \quad (1.4)$$

One sees easily that $P^{\bullet}$ is the simple complex associated with $Q^{\bullet\bullet}$. Let us compute the abutment of the spectral sequence, that is, the homology of $\underline{F}P^{\bullet}$. To this end, using the fact that $L^{\bullet}$ is projective of finite type in every degree, one proves that

$$L^{\bullet} \cong \text{Hom}^{\bullet}(L'^{\bullet}, A^{\bullet}).$$

From the homomorphism $a: A^{\bullet} \rightarrow CA^{\bullet}$, one obtains a homomorphism

$$b: \text{Hom}^{\bullet}(L'^{\bullet}, A^{\bullet}) \longrightarrow \text{Hom}^{\bullet}(L'^{\bullet}, CA^{\bullet}),$$

or equivalently a homomorphism

$$c: L^{\bullet} \longrightarrow P^{\bullet}. \quad (1.5)$$

This being said, it is easy to see, using the fact that $L'^{\bullet}$ is projective of finite type in every degree and bounded on the left, that (1.5) is an injective resolution of $L^{\bullet}$. Using Proposition 1.4, one concludes that $P^{\bullet}$ is homotopy equivalent to $I^{\bullet}$, where $I^{\bullet}$ is the injective resolution of M introduced above in (1.1). It follows that the *abutment* of the first spectral sequence of the double complex $\underline{F}Q^{\bullet\bullet}$, which is $H^*(\underline{F}P^{\bullet})$, is *isomorphic* to $R^*\underline{F}(M)$.

The initial term of the first spectral sequence of the bicomplex $\underline{F}Q^{\bullet\bullet}$ is

$$E_2^{p,q} = {}'H^p({}''H^q(\underline{F}Q^{\bullet\bullet})).$$

For every $p \in \mathbf{Z}$, CA^p is injective. By the hypothesis on $\underline{F}$, the functor (restricted to the category of finitely generated modules)

$$N \longmapsto \underline{F} \text{Hom}(N, CA^p)$$

is exact. Hence one obtains isomorphisms

$${}''H^q(\underline{F} \text{Hom}(L'^{\bullet}, CA^p)) \cong \underline{F} \text{Hom}(H^{-q}(L'^{\bullet}), CA^p).$$

By the definition of $\text{Ext}_{\underline{F}}^*$ as the right-derived functor of $\underline{F} \circ \text{Hom}$, one therefore obtains isomorphisms

$$E_2^{p,q} \cong \text{Ext}_{\underline{F}}^p(H^{-q}(L'^{\bullet}), A).$$

Source note. The French TeX and printed display have $H^q(L'^{\bullet})$ in the last formula. The preceding displayed isomorphism, the statement of Lemma 1.2, and the identity below require $H^{-q}(L'^{\bullet})$. This target therefore uses H^{-q} as a transparent editorial emendation. It is not a restoration from an independent original scan; the French authority remains unchanged pending editorial action.

Now

$$L'^{\bullet} = \text{Hom}^{\bullet}(L^{\bullet}, A^{\bullet}),$$

where $L^{\bullet}$ is a projective resolution of M ; hence there are isomorphisms

$$\text{Ext}^q(M, A) \cong H^q(L'^{\bullet}).$$

Substituting $-q$ in this last identity gives the conclusion. □

2 The Finiteness Theorem

Theorem 2.1⁽¹⁾. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X , and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Suppose that X is locally embeddable in a regular prescheme^{*}. Let $i \in \mathbf{Z}$. Suppose that:*

a) *for every $x \in U = X - Y$, one has*

$$H^{i-c(x)}(\mathcal{F}_x) = 0,$$

where we have set⁽²⁾:

$$c(x) = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \quad (2.1)$$

Then:

⁽¹⁾*Editor's note.* For an analogous statement, but in a somewhat more general situation, see Mme Raynaud (Raynaud M., “Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental,” *Ann. Sci. Éc. Norm. Sup.* (4) **7** (1974), pp. 29–52, Proposition II.2.1).

^{*}This condition may be generalized to the hypothesis that a “dualizing complex,” in the sense defined in R. Hartshorne, *Residues and Duality* (cited in the note to Exposé IV, p. 46), exists locally on X .

⁽²⁾*Editor's note.* As in Exposé V, $H^*(\mathcal{F}_x)$ denotes the local cohomology $H_{\mathfrak{m}_x}^*(\mathcal{F}_x)$.

b) $\mathcal{H}_Y^i(\mathcal{F})$ is coherent.

Corollary 2.2⁽³⁾. Under the hypotheses of the preceding theorem, condition a) is equivalent to:

c) for every $x \in U$ such that $c(x) = 1$, one has $H^{i-1}(\mathcal{F}_x) = 0$.

Corollary 2.3. Let X be a locally Noetherian prescheme that is locally embeddable in a regular prescheme, let Y be a closed subset of X , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, and let n be an integer. The following conditions are equivalent:

- (i) for every $x \in U$, one has $\text{prof } \mathcal{F}_x > n - c(x)$;
- (ii) for every $x \in U$ such that $c(x) = 1$, one has $\text{prof } \mathcal{F}_x \geq n$;
- (iii) for every $i \in \mathbb{Z}$, $\mathcal{H}_Y^i(\mathcal{F})$ is coherent if $i \leq n$;
- (iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$ ⁽⁴⁾.

Assume these results established when X is the spectrum of a regular Noetherian ring A and when $\mathcal{F}$ is the sheaf associated with an A -module of finite projective dimension.

First observe that, if $(X_j)_{j \in J}$ is an open covering of X by open subsets embeddable in a regular scheme, each of the conditions above is equivalent to the conjunction of the analogous conditions obtained by replacing X by X_j , Y by $Y_j = Y \cap X_j$, and $\mathcal{F}$ by $\mathcal{F}|_{X_j}$. Indeed, only the conditions involving $c(x)$ can present a difficulty. Let $x \in U$. If $x \in X_j$, set

$$c_j(x) = \text{codim}(X_j \cap \overline{\{x\}} \cap Y, X_j \cap \overline{\{x\}}).$$

Necessarily $c_j(x) \geq c(x)$. Let $y \in \overline{\{x\}} \cap Y$ be a point that “realizes the codimension,” that is, such that $c(x) = \dim \mathcal{O}_{\overline{\{x\}}, y}$, and let X_j be a member of the covering such that $y \in X_j$. Then $x \in X_j$, and hence $c_j(x) = c(x)$. It follows that condition (a) for the X_j implies condition (a) for X . At this point we have only a partial converse: condition (a) for X implies condition (a) for those X_j such that $c(x) = c_j(x)$, which suffices for our purposes.⁽⁵⁾

Choose a covering of X by open subsets embeddable in a regular prescheme. Applying the preceding observation, we see that we may assume that X is closed in a regular X' . The reduction to X' is then immediate.

We may therefore assume that X is regular, and even affine, by covering X with affine open subsets. That we may assume that $\mathcal{F} = \widetilde{M}$, where M has finite projective dimension, will follow from the next lemma.

Lemma 2.4. Let X be a regular Noetherian prescheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. The function that assigns to each $x \in X$ the projective dimension of $\mathcal{F}_x$ is bounded above.

Indeed, let $x \in X$, and let U be an affine open neighborhood of x . Let $L^\bullet$ be a projective resolution of the module $\mathcal{F}(U)$, where the L^i are finitely generated. By hypothesis, the ring $\mathcal{O}_{X,x}$ is regular, so the projective dimension of $\mathcal{F}_x$ is finite; denote this integer by d . Set

$$K = \ker(L^{-d} \rightarrow L^{-d+1}).$$

The module K_x is free, since d is the projective dimension of $\mathcal{F}_x$ ([M], Ch. VI, Prop. 2.1). By (EGA 0_I, 5.4.1 Errata), it follows that the $\mathcal{O}_U$ -module $\widetilde{K}$ is free on a neighborhood U' of x , with $U' \subset U$. Choose $f \in \mathcal{O}_X(U)$ such that $x \in D(f) \subset U'$. We then have a projective resolution of M_f , where $M = \mathcal{F}(U)$:

$$0 \rightarrow K_f \rightarrow (L^{-d})_f \rightarrow \cdots \rightarrow M_f \rightarrow 0.$$

Source note. After choosing only f , French line 2729 and the compiled page print $M_{f'}$, whereas the displayed resolution ends in M_f . Moreover, the corrected kernel is a submodule of L^{-d} , but French line 2730 prints $(L^{d-1})_f$.

⁽³⁾*Editor's note.* Strictly speaking, this is a corollary of the proof that follows, not of the statement. The implication c) $\Rightarrow$ a) is tautological. The converse is not, but follows from the proof. More precisely, as below, one covers X by open subsets embeddable in regular schemes; as explained below, this permits one to reduce to the case where $X = \text{Spec}(A)$ is affine and regular and $\mathcal{F} = \widetilde{M}$, where M is an A -module of finite projective dimension. In this case it is shown that conditions a) and c) are equivalent to the dual conditions a') and c'). One then shows that c') implies condition d) (see below), which in turn implies a'). See the discussion following 2.4.

⁽⁴⁾*Editor's note.* This condition appeared only in the body of the proof, but not in the statement of the corollary; since it is used in Section 3, it has been added.

⁽⁵⁾*Editor's note.* In fact, condition (a) for X implies condition (a) for every X_j , as asserted in the original text, but establishing this requires reading the proof below in detail. At this point the implication does not appear formal. For a property (a), (b), or (c), append a subscript J to denote its conjunction over the X_j . The proof below establishes $(c_J) \Rightarrow (a_J)$ (this is the chain of implications $(c') \Rightarrow (d) \Rightarrow (a')$). But tautologically $(a) \Rightarrow (c)$, and $(c) \iff (c_J)$; hence $(a) \Rightarrow (a_J)$.

The target uses M_f and $(L^{-d})_f$ as transparent editorial emendations forced by the chosen localization, the kernel definition, and the displayed resolution. These are not restorations from an independent original scan; the French authority remains unchanged pending editorial review.

This proves that the function under consideration is upper semicontinuous. Since X is quasi-compact, the conclusion follows.

We henceforth assume that X is affine, Noetherian, and regular, and that $\mathcal{F} = \widetilde{M}$, where M is a finitely generated A -module, necessarily of finite projective dimension. We shall proceed in several steps. First, we find a condition (d), equivalent to (a), and prove that it is also equivalent to (c). Then, using the spectral sequence of the preceding subsection, we prove $(d) \Rightarrow (b)$. It remains to prove $(iii) \Rightarrow (ii)$; indeed, $(i) \Leftrightarrow (ii) \Rightarrow (iii)$ follows immediately from $(a) \Leftrightarrow (c) \Rightarrow (b)$.

Let $x \in U$. By hypothesis, $\mathcal{O}_{X,x}$ is a regular local ring. If D denotes the dualizing functor for $\mathcal{O}_{X,x}$, it follows from (V, 2.1) that

$$D H^{i-c(x)}(\mathcal{F}_x) \simeq \text{Ext}_{\mathcal{O}_{X,x}}^{d(x)-i}(\mathcal{F}_x, \mathcal{O}_{X,x}),$$

where

$$d(x) = \dim \mathcal{O}_{X,x} + c(x) = \dim \mathcal{O}_{X,x} + \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \quad (2.2)$$

Since X is Noetherian and $\mathcal{F}$ is coherent,

$$D H^{i-c(x)}(\mathcal{F}_x) \simeq (\text{Ext}_{\mathcal{O}_X}^{d(x)-i}(\mathcal{F}, \mathcal{O}_X))_x. \quad (2.3)$$

Moreover, a module vanishes if and only if its dual vanishes (cf. the editor's note (4) on p. 54). For every $q \in \mathbb{Z}$, set

$$\begin{cases} S_q = \text{Supp } \mathcal{E}xt_{\mathcal{O}_X}^q(\mathcal{F}, \mathcal{O}_X), \\ S'_q = S_q \cap U, \quad (U = X - Y), \\ Z_q = \overline{S'_q} \cap Y. \end{cases} \quad (2.4)$$

It follows from formula (2.3) that conditions (a) and (c) are respectively equivalent to:

(a') for every $q \in \mathbb{Z}$ and every $x \in S'_q$, one has $q + i \neq d(x)$;

(c') for every $q \in \mathbb{Z}$ and every $x \in S'_q$, if $c(x) = 1$, then $q + i \neq d(x)$.

Here is condition (d), promised above:

(d) for every $q \in \mathbb{Z}$ and every $y \in Z_q$, one has $q + i \neq \dim \mathcal{O}_{X,y}$.

These conditions are equivalent.

(a') $\Rightarrow$ (c') is recorded for completeness.

(d) $\Rightarrow$ (a'). Indeed, let $q \in \mathbb{Z}$ and $x \in S'_q$. Choose a point $y \in \overline{\{x\}} \cap Y$ that realizes the codimension,⁽⁶⁾ that is, such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}) = c(x). \quad (2.5)$$

Since X is regular at y , it follows that

$$\dim \mathcal{O}_{X,y} = d(x) \quad (\text{cf. (2.2)}). \quad (2.6)$$

But $y \in \overline{\{x\}}$, and hence $y \in Z_q$; the conclusion follows.

(c') $\Rightarrow$ (d). Let $q \in \mathbb{Z}$ and $y \in Z_q$. Assume provisionally that there exists $x \in S'_q$ such that

$$y \in \overline{\{x\}} \quad \text{and} \quad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

(One also says that y is an immediate specialization of x .) It follows that $c(x) = 1$, since y realizes the codimension of $\overline{\{x\}} \cap Y$ in $\overline{\{x\}}$, while $x \notin Y$. Condition (c') therefore gives

$$q + i \neq d(x).$$

Whence the conclusion, upon observing that $d(x) = \dim \mathcal{O}_{X,y}$ by (2.6). The assertion provisionally assumed above is the content of the following lemma.

Lemma 2.5. Let X be a locally Noetherian prescheme and Y a closed subset of X . Set $U = X - Y$, and suppose that U is dense in X . For every $y \in Y$, there exists $x \in U$ such that y is an immediate specialization of x , that is,

$$y \in \overline{\{x\}} \quad \text{and} \quad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

⁽⁶⁾Editor's note. Throughout what follows, closures of points are endowed with the reduced structure.

We have applied the lemma with X equal to the prescheme $\overline{S'_q}$ and Y equal to the subset $Y \cap \overline{S'_q}$.

Proof of Lemma 2.5. There exists $x \in U$ such that $y \in \overline{\{x\}}$. Choose such an x for which $r = \dim \mathcal{O}_{\overline{\{x\}}, y}$ is minimal. We must prove that $r = 1$. By the choice of x , every $z \in \text{Spec}(\mathcal{O}_{\overline{\{x\}}, y})$ with $z \neq x$ lies in Y . Consequently, $\{x\}$ is open in $\text{Spec}(\mathcal{O}_{\overline{\{x\}}, y})$, and the conclusion follows. $\square$

The second step is to deduce (b) from (d).

Set

$$D(Z_q) = \{\dim \mathcal{O}_{X, y} \mid y \in Z_q\}.$$

By (d), for every $q \in \mathbb{Z}$ one has $q + i \notin D(Z_q)$. Applying Proposition VII.2.3, we see that

$$\mathcal{E}xt_Y^{q+i}(\mathcal{E}xt^q(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X) \text{ is coherent.}$$

The initial term of the spectral sequence of the preceding subsection is

$$E_2^{p, q} = \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

It follows that $E_2^{p, q}$ is coherent for every $p, q \in \mathbb{Z}$ such that $p + q = i$. Now there are only finitely many pairs (p, q) such that $p + q = i$,^[S] and this spectral sequence converges to $\mathcal{H}_Y^*(\mathcal{F})$; whence the conclusion.

It remains to prove that (iii) implies (ii). Let

$$i: U \longrightarrow X$$

denote the canonical immersion of U into X . In view of the exact homology sequence associated with the closed subset Y (I, 2.11), one sees that (iii) is *equivalent to*:

(iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$.

French printed p. 94 Indeed, there is an exact sequence

$$0 \longrightarrow \mathcal{H}_Y^0(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_U) \longrightarrow \mathcal{H}_Y^1(\mathcal{F}) \longrightarrow 0.$$

Now $\mathcal{H}_Y^0(\mathcal{F})$ is a quasi-coherent subsheaf of the coherent sheaf $\mathcal{F}$, and is therefore coherent. Hence $\mathcal{H}_Y^1(\mathcal{F})$ is coherent if and only if $i_*(\mathcal{F}|_U)$ is coherent. Moreover, for $p > 0$, the exact cohomology sequence associated with the closed subset Y reduces to isomorphisms

$$R^p i_*(\mathcal{F}|_U) \xrightarrow{\sim} \mathcal{H}_Y^{p+1}(\mathcal{F}).$$

We shall prove that (iv) implies (ii). To this end, recall (ii):

(ii) For every $x \in U$ such that $c(x) = 1$, one has

$$\text{prof } \mathcal{F}_x \geq n.$$

We argue by induction on n .

If $n = 0$, both conditions are vacuous.

If $n = 1$, assume that $i_*(\mathcal{F}|_U)$ is coherent. We argue by contradiction and suppose that there exists $x \in U$ such that $c(x) = 1$ and

$$\text{prof } \mathcal{F}_x = 0,$$

i.e. $x \in \text{Ass } \mathcal{F}_x$. Let $y \in \overline{\{x\}} \cap Y$ be such that

$$\dim \mathcal{O}_{\overline{\{x\}}, y} = 1.$$

Set

$$A = \mathcal{O}_{X, y} \quad \text{and} \quad X' = \text{Spec}(A).$$

Perform the base change $v: X' \longrightarrow X$, which is flat:

$$\begin{array}{ccc} U' = X' \times_X U & \xrightarrow{v'} & U \\ i' \downarrow & & \downarrow i \\ X' & \xrightarrow{v} & X. \end{array} \tag{2.7}$$

^[S]Source note. Taken literally for arbitrary integers, this sentence is false. In context it must be read as concerning the terms that can contribute, in view of the finite-projective-dimension setup above. The French authority is unchanged.

The morphism i is separated, since it is an immersion, and is of finite type, since it is an open immersion and X is locally Noetherian. Since the base change is flat, EGA III, 1.4.15 gives an isomorphism

$$v^*(i_*(\mathcal{F}|_U)) \approx i'_*(v'^*(\mathcal{F}|_U)). \quad (2.8)$$

French printed p. 95 Denote by $\underline{x}$ (respectively, $\underline{y}$) the ideal of A corresponding to x (respectively, y).

Set $\mathcal{G} = v'^*(\mathcal{F}|_U)$. Then $\mathcal{G}$ is coherent and $\underline{x} \in \text{Ass } \mathcal{G}$, so there exists a monomorphism

$$\mathcal{O}_{\overline{\{x\}}} \longrightarrow \mathcal{G};$$

consequently, $i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})$ is coherent. By the choice of y , $\dim(A/\underline{x}) = 1$, and therefore the support of $\mathcal{O}_{\overline{\{x\}}}$ is precisely

$$\overline{\{x\}} = \{x\} \cup \{y\},$$

since $\overline{\{x\}} = \text{Spec}(A/\underline{x})$ as a scheme. It follows that

$$(\mathcal{O}_{\overline{\{x\}}}|_{U'})(U') = \text{Frac}(A/\underline{x}),$$

the field of fractions of $A/\underline{x}$, and

$$i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})(X') = \text{Frac}(A/\underline{x}).$$

But $\text{Frac}(A/\underline{x})$ is not a finitely generated A -module, because $\underline{x}$ is different from the maximal ideal of A . This is a contradiction.

Suppose that $n > 1$, and assume the result established for every $n' < n$. By the induction hypothesis, for every $x \in U$ such that $c(x) = 1$, one has $x \notin \text{Ass } \mathcal{F}_x$. Let x be such a point, and let $y \in \overline{\{x\}} \cap Y$ be an immediate specialization of x , i.e.

$$\dim \mathcal{O}_{\overline{\{x\}}, y} = 1.$$

Perform the base change

$$v: \text{Spec}(\mathcal{O}_{X, y}) \longrightarrow X,$$

retaining the notation of diagram (2.7). Applying EGA III, 1.4.15, one obtains isomorphisms

$$v^*(R^p i_*(\mathcal{F}|_U)) \simeq R^p i'_*(v'^*(\mathcal{F}|_U)), \quad p \in \mathbb{Z}.$$

Thus we are reduced to the case in which X is the spectrum of a local ring A in which $\underline{x}$ is a prime ideal of dimension 1, i.e. $\dim(A/\underline{x}) = 1$. Set

$$\mathcal{F}' = \Gamma_Y(\mathcal{F}) \quad \text{and} \quad \mathcal{F}'' = \mathcal{F}/\mathcal{F}'.$$

We have $\mathcal{F}_x \simeq \mathcal{F}''_x$ and $y \notin \text{Ass } \mathcal{F}''$. Moreover, $\mathcal{F}'|_U = 0$; hence the long exact sequence of the $R^p i_*$ gives isomorphisms

$$R^p i_*(\mathcal{F}|_U) \simeq R^p i_*(\mathcal{F}''|_U), \quad p \in \mathbb{Z}.$$

French printed p. 96 Since $n > 1$, neither x nor y belongs to $\text{Ass } \mathcal{F}''$. Now $\underline{x}$ and $\underline{y}$ are the only prime ideals of A containing $\underline{x}$. It follows from Exposé III, 2.1 that there exists an element $g \in \underline{x}$ that is M -regular, where we have set

$$\mathcal{F} = \widetilde{M}, \quad M = \mathcal{F}(X).$$

Thus there is an exact sequence

$$0 \longrightarrow M \xrightarrow{g'} M \longrightarrow N \longrightarrow 0,$$

in which g' denotes multiplication by g on M . From the resulting long exact sequence one sees that

$$R^p i_*(\widetilde{N}|_U) \quad \text{is coherent for } p < n - 1.$$

Therefore, by the induction hypothesis,

$$\text{prof}(\widetilde{N})_x \geq n - 1, \quad \text{and hence} \quad \text{prof } \mathcal{F}_x \geq n.$$

□

3. Applications

From these results one deduces a coherence condition for the higher direct images of a coherent sheaf under a *morphism that is not proper*.

Theorem 3.1.—Let $f: X \rightarrow Y$ be a morphism of preschemes. Suppose that Y is *locally Noetherian* and that f is *proper*. Suppose that X is *locally embeddable in a regular prescheme*. Let $n \in \mathbb{Z}$. Let U be an open subset of X , and let $\mathcal{F}$ be a *coherent* $\mathcal{O}_U$ -module. Suppose that, for every $x \in U$ such that

$$\text{codim}(\overline{\{x\}} \cap (X \setminus U), \overline{\{x\}}) = 1,$$

one has $\text{prof } \mathcal{F}_x \geq n$. Then the $\mathcal{O}_Y$ -modules

$$R^p(f \circ g)_*(\mathcal{F})$$

are coherent for $p < n$, where g is the canonical immersion of U into X .

Indeed, there is a Leray spectral sequence whose abutment is $R^*(f \circ g)_*(\mathcal{F})$ and whose initial term is given by

$$E_2^{p,q} = R^p f_*(R^q g_*(\mathcal{F})).$$

Moreover, there exists a *coherent* $\mathcal{O}_X$ -module $\mathcal{G}$ such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). It then follows from the preceding section that condition (iv) on p. 74 is satisfied [French printed p. 97], i.e. that $R^q g_*(\mathcal{G}|_U)$ is coherent for $q < n$. Applying the finiteness theorem of EGA III, 3.2.1 to f and the sheaves $R^q g_*(\mathcal{F})$, one finds that $E_2^{p,q}$ is coherent for $q < n$, whence the conclusion.

Proposition 3.2.—Let X be a locally Noetherian prescheme, locally embeddable in a regular prescheme. Let U be an open subset of X , and let $i: U \rightarrow X$ be the canonical immersion. Let $n \in \mathbb{Z}$. Finally, let $\mathcal{F}$ be a coherent Cohen–Macaulay $\mathcal{O}_U$ -module (on $U!$). The following conditions are equivalent:

- (a) $R^p i_*(\mathcal{F})$ is coherent for $p < n$;
- (b) for every irreducible component S' of the closure $\overline{S}$ of the support S of $\mathcal{F}$, one has

$$\text{codim}(S' \cap (X \setminus U), S') > n.$$

Proof, opening.—Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$ -module such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). Applying Corollary 2.3 to $\mathcal{G}$, we find that condition (a) is equivalent to (c):

- (c) for every $x \in S$, one has

$$\text{prof } \mathcal{F}_x > n - c(x), \quad c(x) = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

Proof, (a) $\Rightarrow$ (b).—Indeed, let S' be an irreducible component of $\overline{S}$, and let s be its generic point. Since $\mathcal{F}$ is Cohen–Macaulay, one has $\text{prof } \mathcal{F}_s = \dim \mathcal{O}_{S,s} = 0$. Moreover, $\overline{\{s\}} = S'$, and the conclusion follows.

Proof, (b) $\Rightarrow$ (a), opening.—Let $x \in S$, and let S' be an irreducible component of $\overline{S}$ such that $x \in S'$. Let s be the generic point of S' . Since $\mathcal{F}$ is Cohen–Macaulay, we know that

$$\text{prof } \mathcal{F}_x = \dim \mathcal{O}_{\overline{\{s\}}, x}.$$

Proof, (b) $\Rightarrow$ (a), case split.—If $c(x) = +\infty$, there is nothing to prove. Otherwise, there exists $y \in Y \cap \overline{\{x\}}$ such that

$$c(x) = \dim \mathcal{O}_{\overline{\{x\}}, y}.$$

Proof, (b) $\Rightarrow$ (a), conclusion.—Now $\mathcal{O}_{X,y}$ is a quotient of a regular local ring by hypothesis; hence

$$\dim \mathcal{O}_{\overline{\{s\}}, y} = \dim \mathcal{O}_{\overline{\{s\}}, x} + \dim \mathcal{O}_{\overline{\{x\}}, y} > n.$$

□

Bibliography

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